DHCP Server Configuration

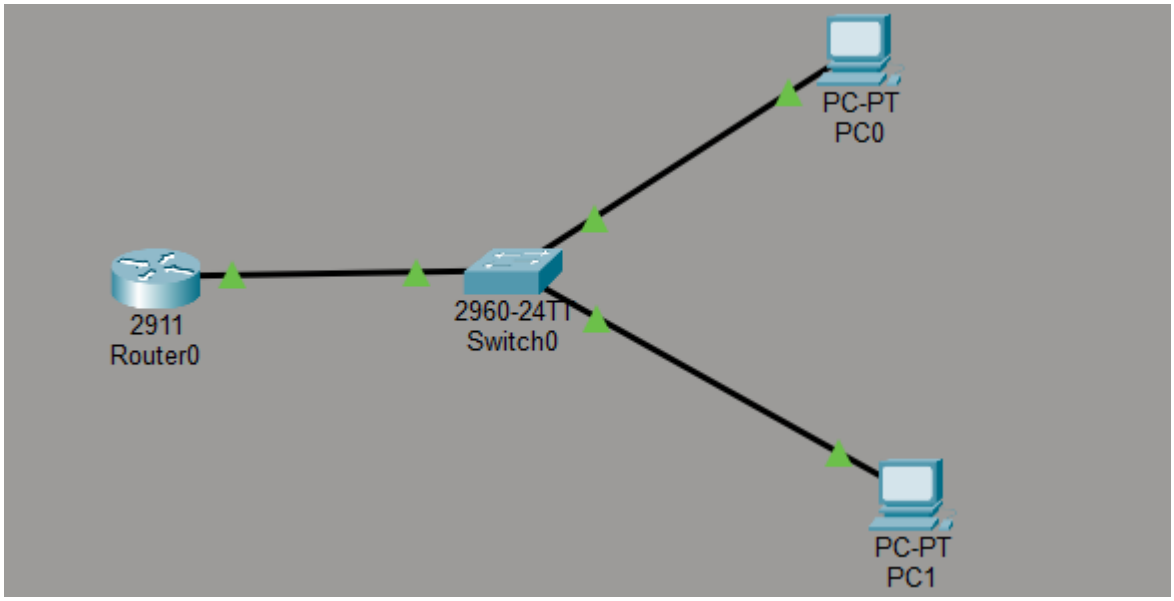
Assignment 4

Objective

To configure a DHCP server on a Cisco router (pool: xyz, network: 192.168.10.0/24, DNS: 1.2.3.4, lease: 3 hours) and verify automatic IP assignment to clients.

Screenshots & Observations

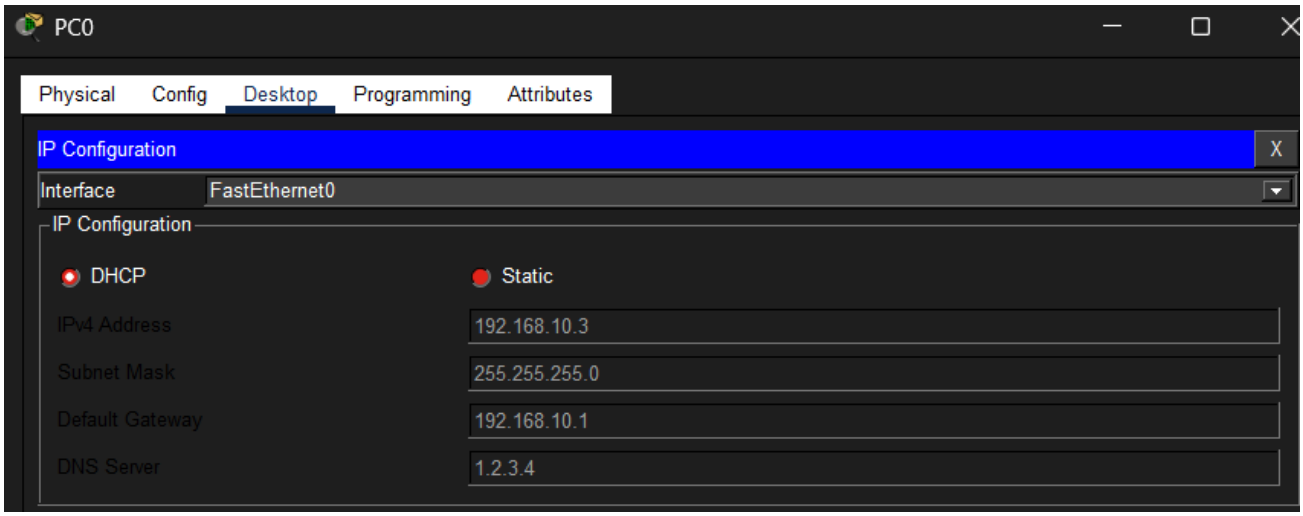
Screenshot 1: Topology Overview



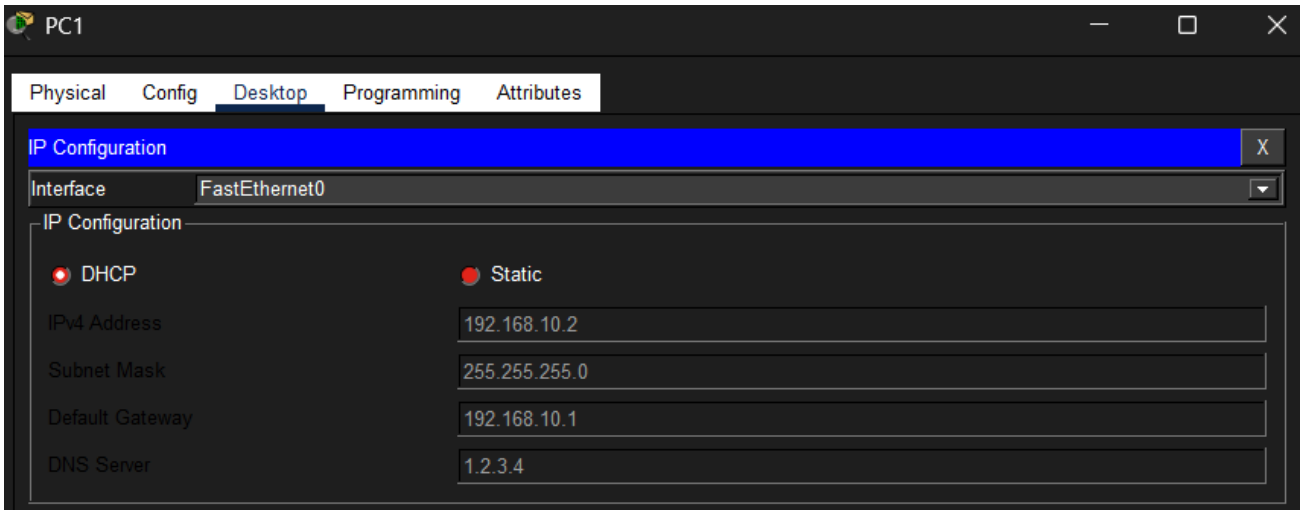
Screenshot 2: Router R1 — DHCP Pool

```
Router#show ip dhcp pool

Pool xyz :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 2
 Excluded addresses             : 1
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.10.1         192.168.10.1     - 192.168.10.254    2    / 1     / 254
```

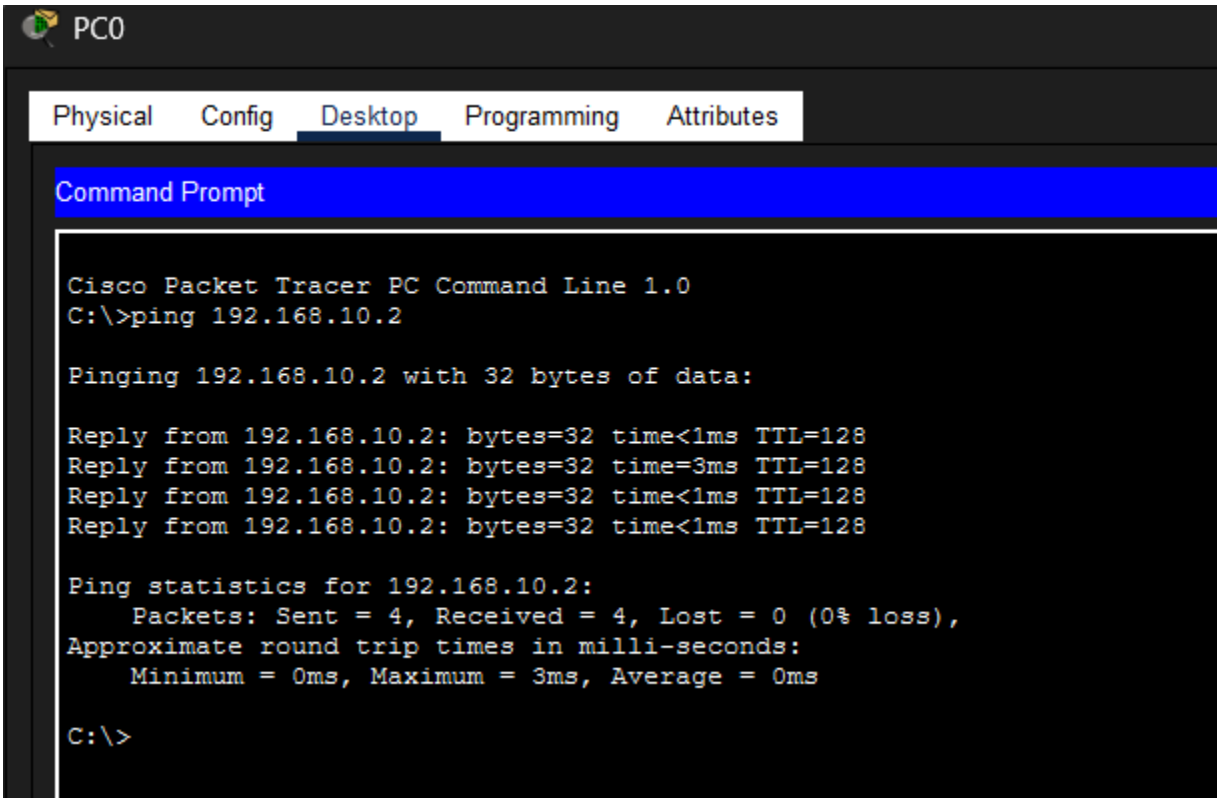
Screenshot 3: PC1 — DHCP Mode IP Address Assigned



Screenshot 5: PC2 — DHCP Mode IP Address Assigned



Screenshot 4: show ip dhcp binding

```
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     00D0.FFDC.B14A           --                     Automatic
192.168.10.3     000D.BD18.B220           --                     Automatic
```

Screenshot 5: Ping Test PC1 to PC2